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Adverse Drug–Drug Interaction Between Phenobarbital and Fluconazole in Two Dogs

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ABSTRACT

Phenobarbital (PB) is an antiseizure medication widely used in dogs that is metabolized by hepatic cytochrome P450 (CYP) enzymes. Fluconazole, a commonly prescribed antifungal medication, inhibits several CYP isoenzymes and can impair PB metabolism. Genetic polymorphisms such as the CYP2C41 gene deletion can alter CYP activity and influence drug interactions, although not well characterized in dogs. We describe two epileptic dogs on chronic PB treatment that developed marked sedation and ataxia, and increased serum PB concentrations after receiving fluconazole. Both dogs were homozygous for the CYP2C41 deletion. Discontinuation of fluconazole resulted in decreased PB concentrations and resolution of clinical signs. These findings suggest fluconazole can inhibit PB metabolism, leading to clinically relevant toxicity, and this interaction does not require CYP2C41 enzyme expression. Monitoring PB concentrations during fluconazole co-administration is advised. Further characterization of the role of CYP enzymes in PB metabolism in dogs is needed to better predict drug interactions.

1 | Introduction

Epilepsy is a common neurological disorder in dogs, with a reported prevalence in the general population ranging from 0.62% to 0.82% [1–3]. This condition is characterized by recurrent unprovoked seizures, defined as the occurrence of at least two epileptic seizures > 24 h apart [4]. Epilepsy can have substantial impact on the quality of life of affected dogs and necessitates long-term management strategies.

Phenobarbital (PB) remains the cornerstone of treatment for dogs with epilepsy because of its rapid absorption and effectiveness in decreasing both the frequency and severity of seizures [5, 6]. However, PB can cause adverse effects such as sedation, ataxia, and liver toxicity, requiring regular monitoring of serum concentrations to ensure appropriate therapeutic dosages and minimize adverse effects [5]. Phenobarbital is primarily metabolized by the liver via the cytochrome P450 (CYP) enzyme system [7], making it susceptible to drug–drug interactions. In people, CYP2C9 and CYP2C19 are the main isoenzymes involved [8]. In dogs, the liver expresses three CYP2C enzymes—CYP2C21, CYP2C41, and CYP2C94 [9, 10], but most dogs lack the CYP2C41 enzyme because of a gene deletion [11, 12]. However, the specific enzymes responsible for PB metabolism in dogs are unknown.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; CNV, copy number variation; CYP, cytochrome P450; GGT, gamma-glutamyl transferase; PB, phenobarbital; RI, reference interval.

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Fluconazole, an antifungal agent, is a potent inhibitor of several CYP isoenzymes, and has been shown to interfere with metabolism of the antiseizure medications phenytoin and carbamazepine in people [13]. Although pharmacokinetic interactions between fluconazole and PB have not been studied, such interactions are possible. Because fluconazole often is prescribed for fungal infections in dogs [14], concurrent administration can occur in epileptic dogs that are on a maintenance treatment regimen that includes PB. In addition, fungal encephalitis can require simultaneous treatment with both fluconazole and PB [15].

We report two dogs that developed signs of PB toxicity with increased serum PB concentrations after initiation of fluconazole treatment, emphasizing a potential drug interaction and the need for monitoring.

2 | Clinical Case Reports

2.1 | Case 1

A 2-year-old male neutered Labrador Retriever with a 5-month history of idiopathic epilepsy was presented to the emergency service at NC State Veterinary Hospital for evaluation of lethargy, difficulty rising, and tetra-ataxia. The dog's treatment regimen included extended-release levetiracetam (80 mg/kg PO q12h), PB (4.0 mg/kg PO q12h) and zonisamide (10.7 mg/kg PO q12h). A serum PB concentration measured 4 weeks after the last dose increase was 31.9 μg/mL, which was within the targeted range of 15–35 μg/mL. One week before presentation, the dog was diagnosed with atopic dermatitis, and culture yielded *Staphylococcus pseudointermedius*, beta-hemolytic *Streptococcus*, and *Trichophyton*. Treatment was initiated with oclacitinib (0.52 mg/kg PO q24h), cefpodoxime (7.1 mg/kg PO q24h), and fluconazole (7.1 mg/kg PO q24h). Within days, the owners noted progressive difficulty rising and reluctance to stand. On physical examination, the dog had focal alopecia and mild erythema along the caudal aspect of the limbs and interdigital spaces. Neurologic examination identified marked proprioceptive tetra-ataxia, with normal cranial nerve function, postural reactions, and spinal reflexes. Laboratory testing identified a serum PB concentration of 79.3 μg/mL (approximately 2.5 times higher than 7 weeks earlier despite no dose change) and a mixed hepatopathy characterized by increased activities of alkaline phosphatase (ALP; 257–768 U/L; reference interval [RI], 9–88), gamma-glutamyl transferase (GGT; 14 U/L; RI, 0–4), and alanine aminotransferase (ALT; 39–52 U/L; RI, 17–78). Suspecting PB toxicity as the cause of the neurological signs, the PB dose was decreased to 1.7 mg/kg PO q12h, targeting a serum concentration of approximately 30 μg/mL. The dosage decrease was calculated based on the formula: (desired serum PB concentration/actual serum PB concentration) × actual PB total daily dosage [16].

The dog was presented again to the neurology service at NC State Veterinary Hospital for reevaluation 3 weeks later. The owners had observed one seizure. Neurologic examination showed improvement in clinical signs, but quiet mentation and mild proprioceptive tetra-ataxia persisted. Laboratory testing performed 1 week earlier by the primary care veterinarian included a normal CBC and serum bile acid concentrations (pre- and post-prandial) and a decreased PB serum concentration of 43.2 μg/mL. However, a serum biochemistry profile identified a progressive mixed hepatopathy, with increases in ALP (818 U/L; RI, 9–88), GGT (22 U/L; RI, 0–4), and ALT (132 U/L; RI, 17–78) activity. Abdominal ultrasound examination indicated no abnormalities in the liver or gastrointestinal tract that could explain the hepatopathy or increased PB concentrations. Dermatologic assessment identified alopecia, mild to moderate erythema, and lichenification on the dorsal surfaces of the digits, carpi, and tarsi, with a repeat dermatophyte culture showing no growth. Given these negative results and concerns about a potential drug interaction between fluconazole and PB, fluconazole was discontinued. Six days after fluconazole discontinuation, the dog was re-presented for cluster seizures, prompting an increase in PB dose (3.2 mg/kg PO q12h). One month later, serum PB concentration was 37.8 μg/mL (approximately 1.14 times lower than the previous concentration). Over the next 5 months, the dog experienced substantial improvement in the tetra-ataxia, sedation and hepatopathy (ALP activity decreased to 764 U/L, and both GGT and ALT activity returned to within the reference interval), and serum PB concentration decreased to 26.0 μg/mL.

2.2 | Case 2

A 6-year-old neutered male Border Collie, with a history of idiopathic epilepsy, was presented to the neurology service at NC State Veterinary Hospital for evaluation of marked sedation and severe non-ambulatory tetra-ataxia. The dog had been receiving extended-release levetiracetam (37 mg/kg PO q12h) and PB (4.4 mg/kg PO q12h; most recent serum PB concentration, 41.6 μg/mL, measured 3 months earlier), and gabapentin (10.3 mg/kg PO q12h). Because of continued seizure activity, zonisamide (10.3 mg/kg PO q12h) was added 2 months earlier. Shortly after, the dog developed pruritus and crusted papules along the dorsum, and dermatologic evaluation identified Malassezia overgrowth and superficial pyoderma. Treatment was initiated with fluconazole (5.1 mg/kg PO q24h) and 2% chlorhexidine shampoo. Concern for a possible cutaneous adverse drug reaction to zonisamide, combined with persistent seizure activity, prompted initiation of potassium bromide (50 mg/kg loading dose PO q12h for 5 days, then 30 mg/kg PO q24h) and a 20-day tapering of zonisamide. One week later, the dog was presented to the neurology service for evaluation. Physical examination identified persistent crusted papules on the dorsum and saliva-stained fur on the hind paws. Neurologic examination identified marked sedation, inability to stand, and severe non-ambulatory tetra-ataxia with absent proprioceptive positioning and hopping responses in all limbs, whereas cranial nerves and spinal reflexes were normal. Laboratory analysis identified markedly increased PB concentration (66.5 μg/mL) and a mixed hepatopathy with changes in ALP (149–296 U/L; RI, 9–88), GGT (6–7 U/L; RI, 0–4), and ALT (275–204 U/L; RI, 17–78) activity compared to laboratory testing performed 15 weeks previously.

The dog was hospitalized for IV fluid therapy (0.45% NaCl with KCl supplementation). Fluconazole and gabapentin were discontinued, and the PB dose was temporarily decreased to 1.7 mg/kg PO q12h and then increased back to 3.8 mg/kg PO q12h after 2 days. At the time of discharge on day 3, the dog was

quiet and alert, with moderate tetra-ataxia and delayed to absent proprioceptive positioning and hopping responses in all limbs. On reevaluation 17 days later, the owner reported that the dog was approximately 85% back to normal, with a good appetite and decreased sedation, but intermittent scuffing of the hind limbs was noted. At this time, the serum PB concentration was 27.2 μg/mL, despite the dose increase.

3 | CYP2C41 Gene Deletion Polymorphism

Because dogs can have three CYP2C enzymes, but most dogs lack the CYP2C41 isoform [11], EDTA-preserved whole blood samples from both dogs were analyzed to determine presence of the CYP2C41 gene deletion. Samples were genotyped for the *CYP2C41* copy number variation (CNV) polymorphism using a quantitative real-time PCR–based assay as described previously [17]. Briefly, predesigned TaqMan gene expression assays (Applied Biosystems, Thermo Fisher Scientific, Waltham, MA) were used to amplify a 103-base pair region within exon 8 of the *CYP2C41* gene, with the canine *UGT1A* gene (exon 5) serving as the reference. Paired reactions were run using TaqMan Gene Expression Master Mix on a Bio-Rad CFX96 Real-Time System (Bio-Rad, Hercules, CA). The cycling conditions included enzyme activation at 95°C for 10 min, followed by 70 cycles of 95°C for 15 s and 60°C for 1.5 min. Gene copy number was determined using the $2^{-\Delta\Delta Ct}$ method. Both dogs had the *CYP2D41* CNV −/− genotype indicating the absence of the *CYP2C41* gene.

4 | Discussion

Our report highlights a potential drug interaction between PB and fluconazole in two dogs. Both dogs had clinically relevant increases in serum PB concentrations after the addition of fluconazole, resulting in signs of sedation, ataxia, and increases in liver enzyme activities consistent with PB toxicity. Serum PB concentrations decreased after fluconazole discontinuation, supporting the premise that fluconazole inhibited PB metabolism in the dogs, leading to drug accumulation. Although fluconazole-induced hepatotoxicity impairing PB clearance is theoretically possible, the rapid clinical and biochemical recovery after discontinuation makes this explanation unlikely. Thus, CYP inhibition remains the most plausible explanation for the increased PB concentrations observed.

In humans, fluconazole inhibits CYP enzymes, particularly CYP2C19 and to a lesser extent CYP2C9 and CYP3A4 [18], which can affect the metabolism of various medications [19]. Fluconazole can increase plasma PB concentrations by decreasing the metabolism of PB by CYP2C19 and CYP2C9 [20, 21]. The specific CYP enzymes that fluconazole inhibits in dogs are not fully known, but the inhibition of CYP enzymes raises the potential for increased drug concentrations and an increased risk of adverse effects or toxicity. Other medications that can be affected by fluconazole's inhibition of the CYP system in dogs include cyclosporine, leading to potential nephrotoxicity [22], anesthetic drugs (ketamine and midazolam) causing prolonged anesthesia duration [23], and opioid analgesic drugs (tramadol and methadone) increasing central opioid effects [24, 25]. Furthermore, genetic polymorphisms in CYP enzymes can alter PB clearance in humans [26, 27]. The main isoenzymes involved in PB metabolism in humans are CYP2C9 and CYP2C19. Variants in both loss and gain of function can determine metabolic phenotypes that range from poor to ultra-rapid metabolizers and consequently affect drug concentrations and therapeutic outcomes [7].

Although the specific enzymes involved in PB metabolism in dogs have not been identified, species-specific differences in CYP isoenzymes exist because of genetic variability [7]. Nonetheless, orthologous enzymes (derived from a common ancestral gene) are present across species and often exhibit similar substrate specificities [7], making it plausible that CYP2C enzymes contribute to PB metabolism in dogs. Hepatic expression of CYP2C21, CYP2C94, and CYP2C41 has been described in dogs [10], but most dogs lack the CYP2C41 enzyme because of a gene deletion [11, 17]. This genetic variability, coupled with breed-associated differences in CYP expression, can influence drug metabolism and therapeutic responses [7, 28].

Both dogs reported here had the *CYP2C41* deletion. Consequently, we can conclude that the presence of the *CYP2C41* gene is not required for the fluconazole-PB interaction in dogs. However, we cannot determine whether the interaction would differ in dogs that express this enzyme. Dogs possessing *CYP2C41*, particularly individuals with two gene copies, have been shown to possess abundant hepatic *CYP2C41* enzyme, and recombinant CYP2C41 exhibits higher catalytic activity toward certain substrates compared to other CYPs in dogs [17]. If CYP2C41 plays a role in PB metabolism, its presence could decrease the extent of fluconazole's inhibitory effects, potentially mitigating toxicity. Whether this is the case or not also would depend on whether fluconazole inhibits CYP2C41. Additional investigation is warranted to clarify whether *CYP2C41* contributes to PB clearance and whether its absence might increase susceptibility to adverse drug interactions in affected dogs.

Effective seizure management in dogs receiving long-term PB necessitates vigilance regarding potential pharmacological interactions, particularly with fluconazole. This antifungal agent, frequently prescribed for systemic infections or fungal meningoencephalitis, acts as a CYP enzyme inhibitor and can markedly increase serum PB concentrations, thereby increasing the risk of toxicity. A thorough understanding of PB metabolism and the modulatory effects of CYP inhibition is paramount to optimizing therapeutic outcomes. Consequently, clinicians should obtain a baseline serum PB concentration before initiating fluconazole treatment, followed by reassessment 7–10 days after starting treatment, with close monitoring of clinical signs to promptly detect and mitigate potential toxicity using judicious dosage adjustments.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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